HW 11

8.10

Let X_1, X_2 be independent random variables, all with the same distribution $P(X_i = 0) = \frac{1}{2}, P(X_i = 1) = \frac{1}{3}, P(X_i = 2) = \frac{1}{6}$

Let $S = X_1 + X_2$. Use the moment generating function of S to find the probability mass function of S.

✓ Answer

$M(t) = \frac{1}{2}e^{0t} + \frac{1}{3}e^{t} + \frac{1}{6}e^{2t}$

$M_S(t) = \frac{1}{4}e^{0t} + \frac{1}{3}e^{t} + \frac{5}{18}e^{2t} + \frac{1}{9}e^{3t} + \frac{1}{36}e^{4t}$

$$P(S = s) = \begin{cases} \frac{1}{4} & 0 \\ \frac{1}{3} & 1 \\ \frac{5}{18} & 2 \\ \frac{1}{9} & 3 \\ \frac{1}{36} & 4 \end{cases}$$

8.15

Let (X, Y) be a uniformly distributed random point on the quadrilateral D with vertices $(0,0)$, $(2,0)$, $(1,1)$, and $(0,1)$. Calculate the covariance of X and Y. Based on the description of the experiment, should it be negative or positive?

✓ Answer

It should be negative since theyre slightly negatively correlated

$$E[X] = \frac{2}{3}\int_0^1 \int_0^{-y+2} x\,dx\,dy = \frac{2}{3}\int_0^1 \frac{y^2}{2} - 2y + 2\,dy = \frac{2}{3}\left(\frac{1}{6} - 1 + 2\right) = \frac{7}{9}$$

$$E[X] = \frac{2}{3}\int_0^1 \int_0^{-y+2} y\,dx\,dy = \frac{2}{3}\int_0^1 2y - y^2\,dy = \frac{2}{3}\left(1 - \frac{1}{3}\right) = \frac{4}{9}$$

$$E[XY] = \frac{2}{3}\int_0^1 \int_0^{-y+2} xy\,dx\,dy = \frac{2}{3}\int_0^1 \frac{y^3}{2} - 2y^2 + 2y\,dy = \frac{1}{3}\left(\frac{1}{4} - \frac{4}{3} + 2\right) = \frac{11}{36}$$

$$\mathrm{Cov}(X, Y) = \frac{11}{36} - \frac{7}{9}\frac{4}{9} = -\frac{13}{324}$$

8.26

Suppose I have an urn with 9 balls: 4 green, 3 yellow, and 2 white ones. I draw a ball from the urn repeatedly with replacement.

a

Supposed I draw n times. Let X_n be the number of time I saw a green ball followed by a yellow ball. Calculate the expectation $E[X_n]$.

✓ **Answer**

By independence, any two draws will have a probability of green than yellow of $\frac{4}{27}$.

There are $n-1$ valid adjacent pairs when drawing n balls.

$E[X_n] = (n-1)\frac{4}{27}$

b

Let Y be the number of times I drew a green ball before the first white draw. Calculate $E[Y]$. Can you give an intuitive explanation for your answer?

✓ **Answer**

The number of draws until a white ball will be $\text{Geom}\left(\frac{2}{9}\right)$ which means that we expect there to be $\frac{9}{2}$ balls drawn until a white.

Each of those balls have a change of $\frac{4}{9}$ of being green. This follows a $\text{Binom}\left(\frac{9}{2}, \frac{4}{9}\right)$, which has an expectation of 2.

8.37

A cereal company announces a collection of 10 different toys for prizes. Every box of cereal contains two different randomly chosen toys from the series of 10. Let X be the number of different types of toys I have accumulated after buying 4 boxes of cereal.

a

Find the mean of X.

✓ **Answer**

The probability of getting toy i after N draws would be $1 - \frac{4}{5}^N$

So our expected number of toys for 10 different toys with 4 pulls would be $10\left(1 - \frac{4}{5}^4\right) = \frac{738}{125}$

b

Find the variance of X.

✓ **Answer**

Since X is the sum of 10 indicators, its variance is the sum of the variances + 2 of each covariance.

$$\mathrm{Var}(I) = \frac{4}{5}^N \left(1 - \frac{4}{5}^N\right) = \frac{94464}{390625}$$

9.6

Nate is a competitive eater specializing in eating hot dogs. From past experience we know that it takes him on average 15 seconds to consume hot dog, with a standard deviation of 4 seconds. In this year's hot dog eating contest he hopes to consume 64 hot dogs in just 15 minutes. Use the CLT to approximate the probability that he achieves this feat of skill.

✓ **Answer**

By CLT, this distribution is roughly normal.
T = seconds to eat 64 hot dogs.
$T \approx \mathrm{Norm}(64 \cdot 15, (\sqrt{64} \cdot 4)^2)$

$$P(T \leq 900) = P\left(Z < \frac{900-960}{32}\right) \approx 0.03$$

9.21

Let $X_1, \ldots, X_{500}$ be i.i.d. random variables with expected value 2 and variance 3. The random variables $Y_1, \ldots, Y_{500}$ are independent of the X_i variables, also i.i.d., but they have expected value 2 and variance 2. Use the CLT to estimate $P(\sum_{i=1}^{500} X_i > \sum_{i=1}^{500} Y_i + 50)$.

Hint: Use the CLT for the random variables $X_i - Y_i$

✓ **Answer**

$Z_i = X_i - Y_i \sim N(0, 5)$
$Z_\mu = \sum^{500} Z_i = N(0, 50^2)$
$P(Z_\mu > 50) = P(Z > 1) = 0.1587$